

Resolving a Conjecture on Stack-Sorting

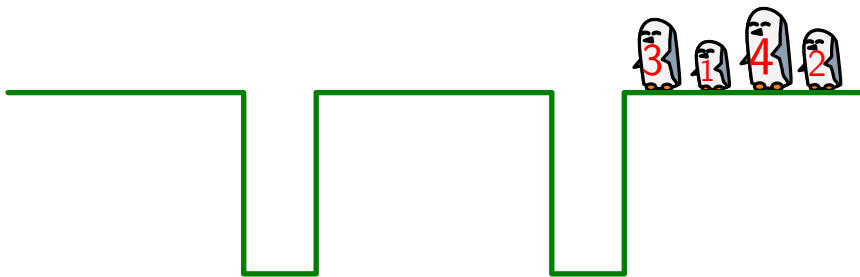
Michael Luo
Research Science Institute
Mentor: Ryota Inagaki

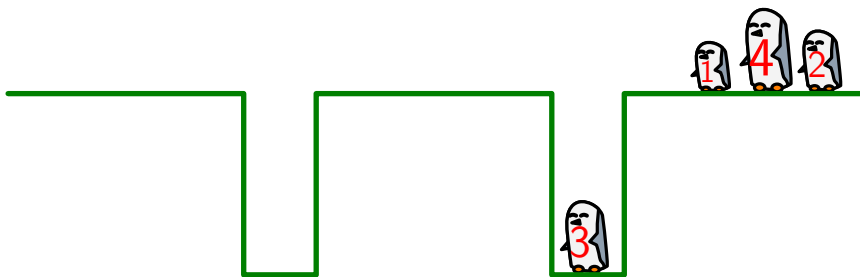
August 2026

Penguins

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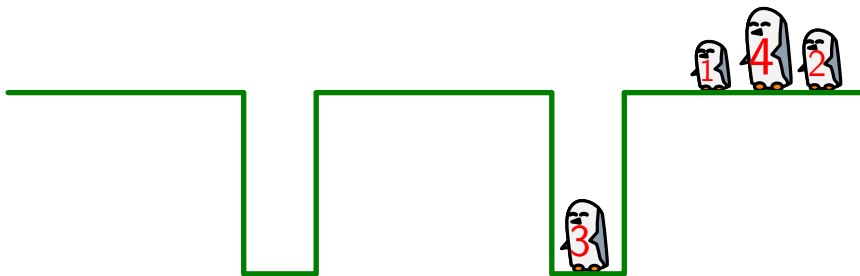
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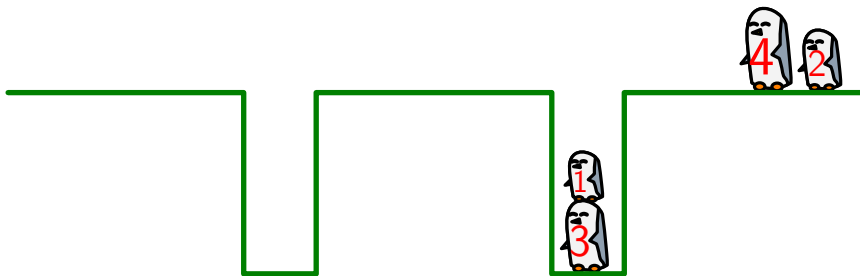
Penguins are

- **Impatient:** small penguins will always jump on big penguins



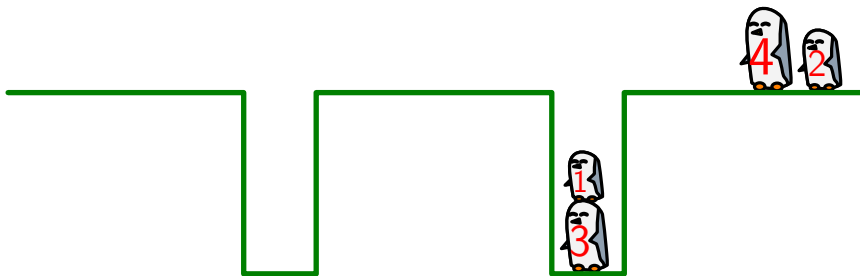
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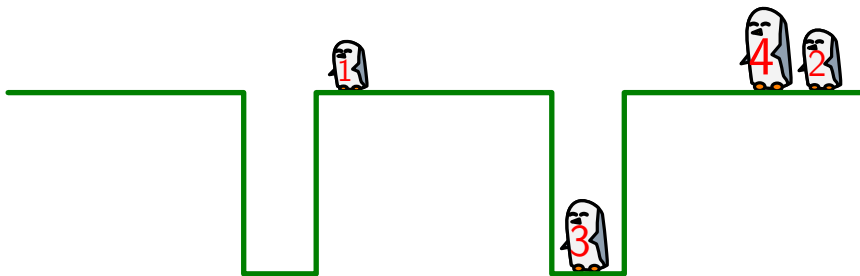
Penguins are

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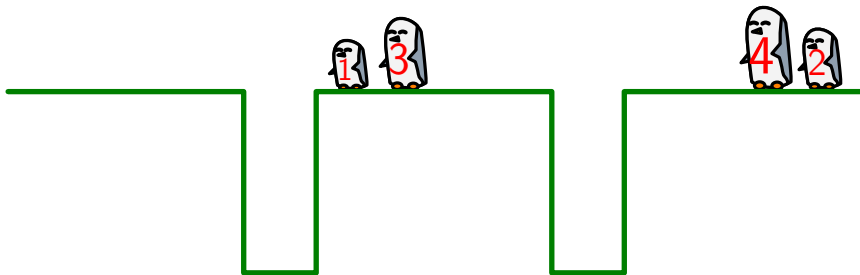
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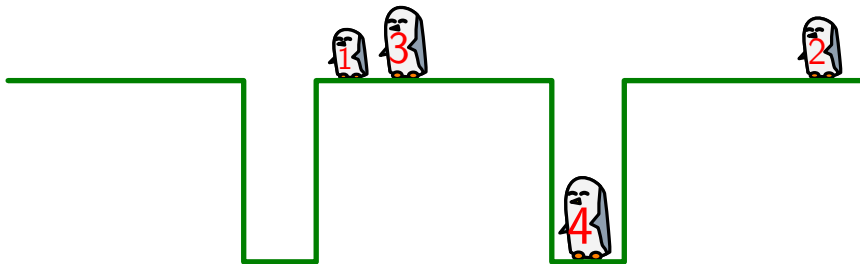
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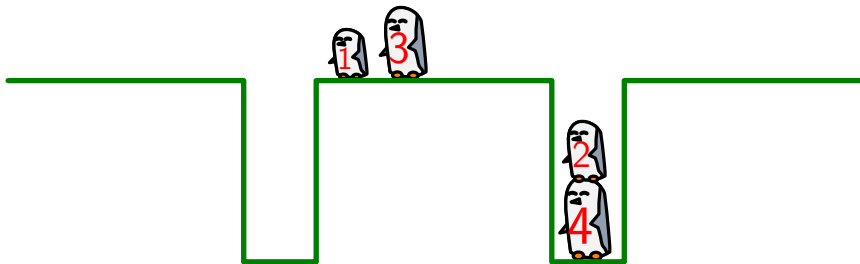
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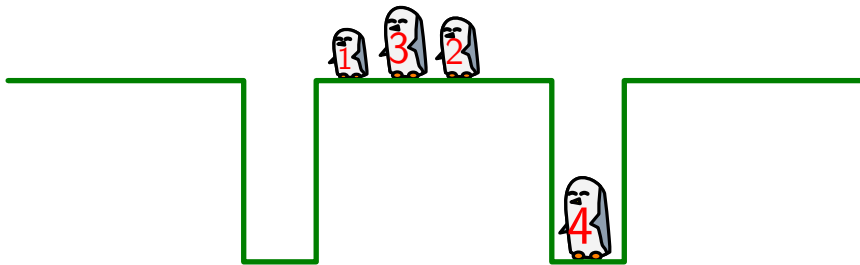
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Penguins by Evan Chen, used with permission. Figure generated with AI assistance.

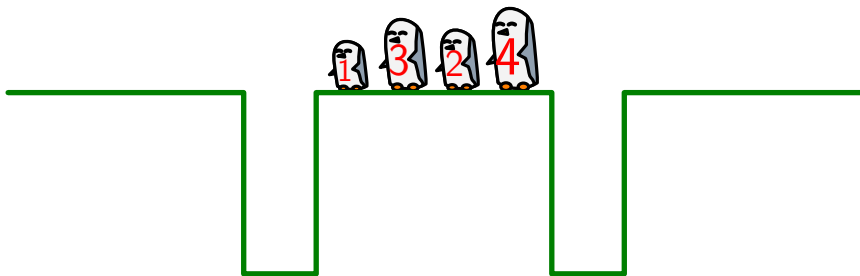
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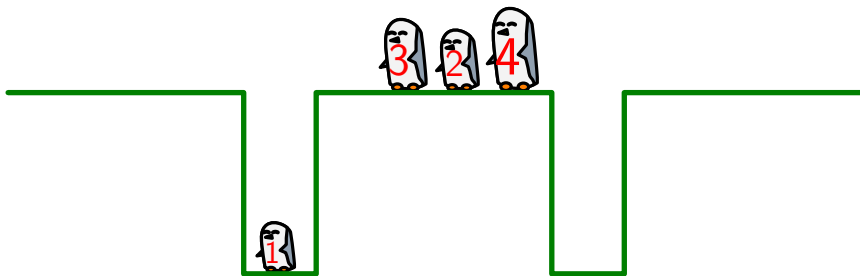
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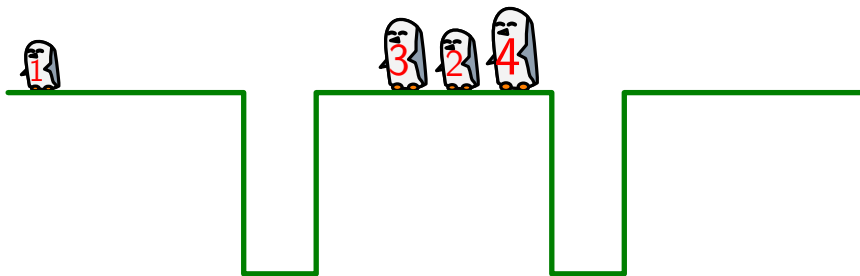
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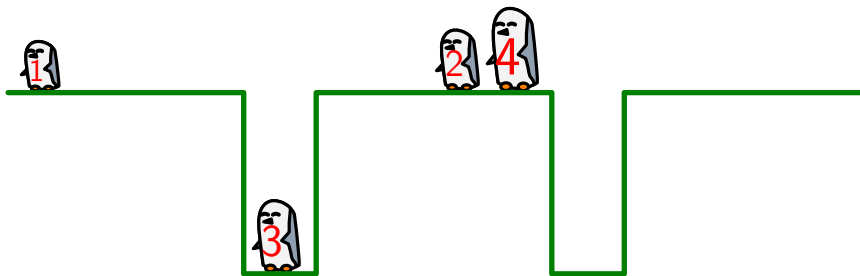
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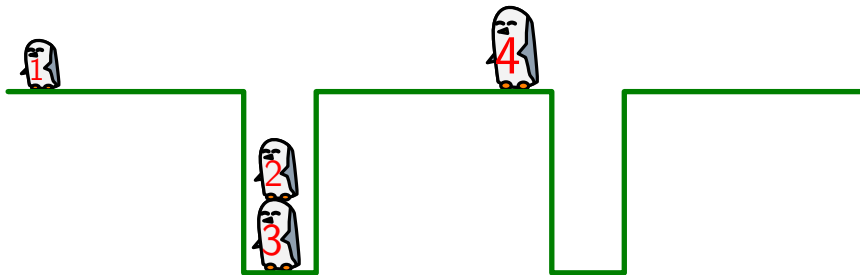
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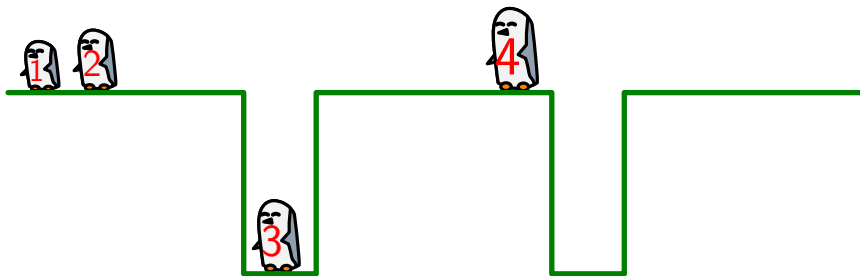
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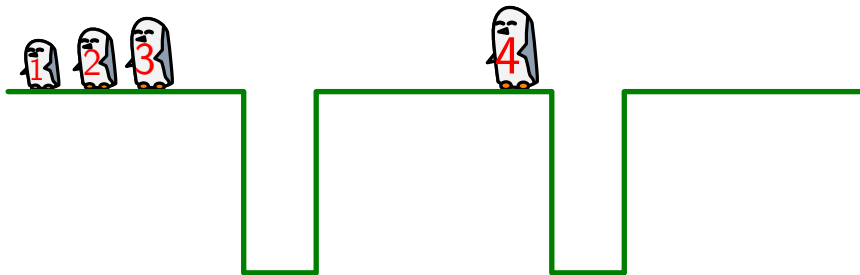
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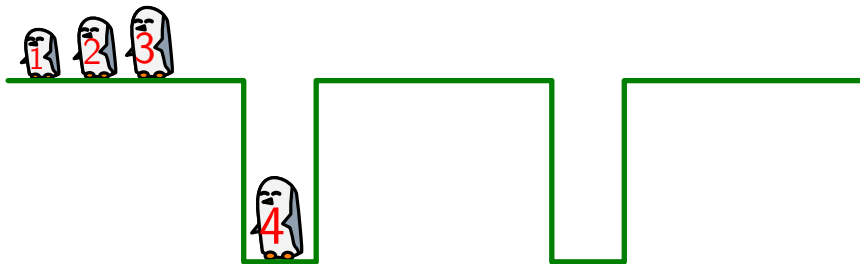
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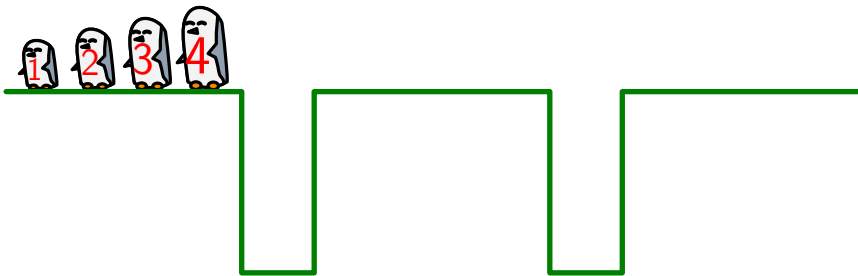
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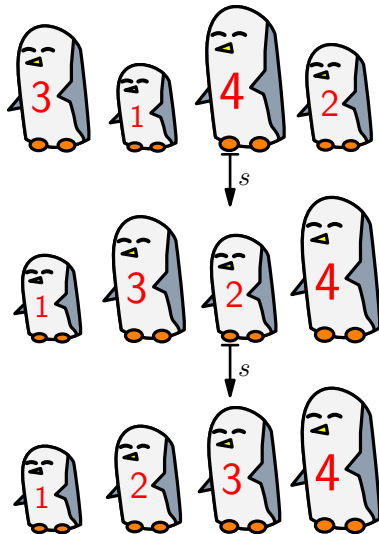
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In notation: $s(3142) = 1324$, $s(1324) = 1234$

Main result

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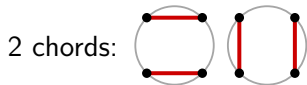
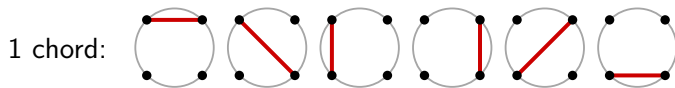
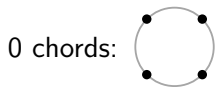
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Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

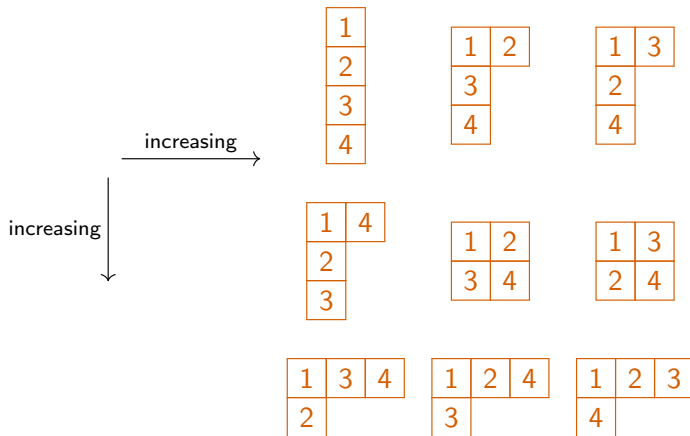
n th Motzkin counts:

Ways to draw any # of disjoint chords connecting n points on a circle



n th Motzkin counts:

Number of **standard Young tableaux** with width ≤ 3 (Regev 1981)



nth Motzkin counts:

Number of 4321-, (3412,2413)-, (3412,3241) and 3412-avoiding involutions in S_n .
Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any two consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
From [David Callan](#), Jul 15 2004: (Start)
Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $X \times Y$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.
Number of Dyck n -paths with no UUU. (Given such a Dyck n -path, change each UUU to UDU , then change each remaining UUD to F . This is a bijection to Motzkin n -paths. Example with $n=5$: $UUDUUDUDD \rightarrow UUFUDDU$.)
Number of Dyck $(n+1)$ -paths with no UUU. (Given such a Dyck $(n+1)$ -path, mark each U that is followed by a D and each D that is not followed by a U. Then change each unmarked U with each matching D that is marked to an F . Lastly, delete all the marked steps. This is a bijection to Motzkin n -paths. Example with $n=6$ and marked steps in small type: $UUDUUDUDDU \rightarrow UUDUUDUDDU \rightarrow UUDUUDUDDU \rightarrow UUDUUDUDDU$.
 $a(n)$ is the number of strings of length $2n+2$ from the following recursively defined set: L contains the empty string and, for any strings a and b in L , we also find (ab) in L . The first few elements of L are $e, (1,0), ((1,1)), ((1,0),(1,0)), ((1,0),(1,1)), ((1,1),(1,0))$. The proof that $a(n)$ is less than or equal to $C(n)$, the n -th Catalan number. See [Orinick link](#) (2024). - [Saul Schleier](#)
[sausch(AT)math.rutgers.edu, Feb 23 2006 (Additional Linked content added by [William P. Orrick](#), Jun 13 2024).]
 $a(n)$ = number of Dyck n -paths all of whose valleys have even x -coordinate (when path starts at origin). For example, $T(4,2)=3$ counts $UUUUUUUU$, $UUUUUUUU$, $UUUUUUUU$. Given such a path, split it into n subpaths of length 2. Each subpath is of type UD or DU . $UD \rightarrow U$ and $DU \rightarrow D$. So the path $UUUUUUUU$ becomes $UUUUUUUU$.
 x -coordinate. This is a bijection to Motzkin n -paths. - [David Callan](#), Jun 07 2006
Also the number of standard Young tableaux of height ≤ 3 . - [Mike Zabrocki](#), Jun 24 2007
 $a(n)$ is the number of RNA shapes of size $2n+2$. RNA shapes are essentially Dyck words without "directly nested" motifs of the form $A1B1B1C$, for A, B and C Dyck words. The first RNA shapes are $\{(); (1,1); ((1,1), (1,1)); ((1,1),(1,1)), ((1,1),(1,1)), ((1,1),(1,1)), ((1,1),(1,1))\}$ - [Yann Ponty](#) (ponty(AT)iri.fr), May 30 2007
The sequence is self-generated from top row A going to the left starting $(1,1)$ and bottom row B , the same sequence but starting $(0,1)$ and going to the right. Take dot product of A and B and add the result to n -th term of A to get the $(n+1)$ -th term of A . Example: $a(5) = 21$ as follows: Take dot product of $A = (0, 4, 2, 1, 1)$ and $B = (1, 1, 2, 4, 0)$, $0 + 1 = 1$, $4 + 2 = 6$, $2 + 4 = 6$, $1 + 0 = 1$, $1 + 0 = 1$.
Equals $A005737 + A080573$ shifted (i.e., $(1,2,3,13,35,96,...) \rightarrow (1,1,2,5,13,35,96,...)$). - [Gary W. Adamson](#), Dec 28 2008
Starting with offset 1 iterates of $M = (1,1,8,0,...)$, where M is a triangular matrix with $(0,1,1,...)$ in the main diagonal and $(1,1,1,...)$ in the super and subdiagonals. - [Gary W. Adamson](#), Jan 07 2009
 $a(n)$ is the number of involutions of $\{1,2,...,n\}$ having genus 0. The genus $g(p)$ of a permutation p of $\{1,2,...,n\}$ is defined by $g(p) = (n+1 - \text{cyc}(p))/2$, where p is the inverse permutation of p , c is $\text{cyc}(p)$, $\text{cyc}(p)$ is the number of cycles of the permutation p . Example: $a(4) = 9$. Indeed, $p = 3412 = (13)(24)$ is the only involution of $\{1,2,3,4\}$ with genus 0. This follows easily from the fact that a permutation p of $\{1,2,...,n\}$ has genus 0 if and only if the cycle decomposition of p gives a noncrossing partition of $\{1,2,...,n\}$ and each cycle of p is increasing (see Lemma 2.1 of the Buluce-Sunior reference). [Also, redundantly, for $p = 3412 = (13)(24)$ we have $c(p) = 23413412 = 4123 = (1432)$ and so $g(p) = (1/2)((4+1)-1) = 1$. - [Emeric Deutsch](#), May 21 2018
Let w_i, j, n denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w_i, j, n = w_i, j, n-1 + w_i-1, j, n-1 + w_i-1, j, n-1 + w_i+1, j-1, n-1$ with boundary conditions $w(0,0,1) = 1$ and $w_i, j, n = 0$ if $i < 0$ or $j < 0$ or $n < 0$. Then $a(n) = \sum_{i=0}^n w_i, i, n = \sum_{i=0}^n w_i, j, n$, w_i, j, n is the number of such walks of length n . - [Peter Luschny](#), May 21 2018
 $a(n)/n!$ tends to 3/8 as $N \rightarrow \infty$: $(1+2\cos(2\pi/3))$ relating to Longest odd N regular polygon diagonals, by way of example, $n=7$: Using the tridagonal generator [cf. comment of Jan 07 2009], for $n=7$ (from $n=7$), we extract $n=7$ of the 3×3 matrix, $(0,1,0), (1,1,1), (0,1,1)$ with an eval of 2.24697... the longest Heptagon diagonal with edge 1. As N tends to infinity, the diagonal lengths tend to 3/8, the convergent of the sequence. - [Gary W. Adamson](#), Jun 08 2011
Number of $(n+1)$ -permutations avoiding the pattern 132 and the dotted pattern 231dot(1). - [Jean-Luc Baril](#), Mar 07 2012
Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix x of w we have $\#(x,a) \geq \#(x,b)$ and $\#(x,b) \geq \#(x,c)$, where $\#(x,i)$ is the letter i in word x . For $n=4$: 9 words are: $aaaa, aaaa, aaba, aabb, aabb, aabb, aabb, aabb, aabb$. - [Alois P. Heinz](#), May 26 2012
Number of length- n restricted growth strings (RGS) $(r(1), r(2), ..., r(n))$ such that $r(1) \leq r(k) \leq k$ and $r(k) \leq r(k-1)$; for example, the 9 RGS for $n=4$ are 1010, 1012, 1201, 1210, 1212, 1230, 1232, 1234, 1234. - [Joerg Arndt](#), Apr 20 2013
Number of length- n restricted growth strings (RGS) $(r(1), r(2), ..., r(n))$ such that $r(1) \leq r(k) \leq k$ and $r(k) \leq r(k-1) \leq 1$; for example, the 9 RGS for $n=4$ are 0000, 0002, 0003, 0004, 0022, 0024, 0032, 0222, 0224. - [Joerg Arndt](#), Apr 17 2013
Number of (4231,5276143)-avoiding involutions in S_n . - [Alexander Burstein](#), Mar 05 2014
 $a(n)$ is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A258888](#).
[Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)
A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection, [A080186](#) (empirical observation).
A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}(k=0,n) k! \text{Fibonacci}(2k)$, offset 3, [A197499](#) (empirical observation). (End)
Conjecture: $(2/n) \text{Sum}(k=1,n) (2k+1)k!k^2$ is an integer for each positive integer n . - [Zhi-Wei Sun](#), Nov 16 2012
The Rubey and Stump reference proves a refinement of a conjecture of René Marczinkin, which they state as: "The number of 2-Gorenstein algebras which are Nakayama algebras with n simple modules and have an orientation line as associated quiver equals the number of Motzkin paths of length n ." - [Eric M. Schmidt](#), Dec 16 2017
Number of U -(k)-equivalence classes of lukasiewicz paths. lukasiewicz paths are P -equivalent if the positions of pattern p are identical in these paths. [Sergey Kirilov](#), Apr 08 2018
If tau_1 and tau_2 are two distinct permutation patterns chosen from the set $\{(32,231,312)\}$, then $a(n)$ is the number of valid hook configurations of permutations of $\{1, n\}$ that avoid the patterns tau_1 and tau_2 . - [Colin Defant](#), Apr 28 2019
Number of permutations of length n that are sorted to the identity by a consecutive-321-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020
From [Helmut Prodinger](#), Dec 13 2020: (Start)
The number of paths in the first quadrant starting at $(0,0)$ and consisting of n steps from the set $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$.
infinite set $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$.
For example, denoting $U(1,1), D(1,-1), D_2(1,-2)$ for $j = 2$, $a(4)$ counts $UUUU, UUUD, UUUD_2, UUUD_3, UUUD, UUUD, UUUD, UUUD$.
This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000.
See Prodinger link that contains a bijection to Motzkin paths. (End)
Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1908-1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Asirfan Fikder](#), Apr 15 2021
Number of Motzkin n -paths $a(n)$ is split into [A017582\(n\), number of even Motzkin \$n\$ -paths, and \[A434386\\(n\\), number of odd Motzkin \\$n\\$ -paths. The value \\[A017582\\\(n\\\) - A434386\\\(n\\\) can be called the "shadow" of \\\$a\\\(n\\\)\\\$ \\\(see \\\[A434373\\\]\\\(#\\\)\\\). - \\\[Gennady Eremin\\\]\\\(#\\\), May 17 2021
Conjecture: if \\\$p\\\$ is a prime of the form \\\$6m+1\\\$ \\\(\\\[A002476\\\\), then \\\\$a\\\\(p-2\\\\)\\\\$ is divisible by \\\\$p\\\\$. Currently, no counterexample exists for \\\\$p < 10^7\\\\$. Personal communication from \\\\[Robert Gerbicz\\\\]\\\\(#\\\\): mod such \\\\$p\\\\$ this is equal to \\\\$a\\\\(6p-7\\\\) \pmod{p}\\\\$ with comment: "Every \\\\[A066796\\\\\(n\\\\\) from \\\\\[A066796\\\\\\(p-1\\\\\\)/2 is divisible by prime \\\\\\$p\\\\\\$ of form \\\\\\$6m+1\\\\\\$ ". - \\\\\\[Serge Bateman\\\\\\]\\\\\\(#\\\\\\), Feb 08 2022
From \\\\\\[Rob Burns\\\\\\]\\\\\\(#\\\\\\), Nov 11 2024: \\\\\\(Start\\\\\\)
The number of paths is proved to be equal to the number of paths by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that \\\\\\$a\\\\\\(p-2\\\\\\) \equiv 0 \pmod{p}\\\\\\$ when \\\\\\$p \equiv 1 \pmod{6}\\\\\\$ and \\\\\\$a\\\\\\(p-2\\\\\\) \equiv -1 \pmod{p}\\\\\\$ when \\\\\\$p \equiv -1 \pmod{6}\\\\\\$.
In fact, the 2012 paper by Burns establishes several general congruences for \\\\\\$a\\\\\\(p^k - 2\\\\\\)\\\\\\$ where \\\\\\$k \geq 1\\\\\\$.
If \\\\\\$p \equiv 1 \pmod{6}\\\\\\$ then \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv 0 \pmod{p}\\\\\\$ for \\\\\\$k \geq 1\\\\\\$.
If \\\\\\$p \equiv -1 \pmod{6}\\\\\\$ then \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv -1 \pmod{p}\\\\\\$ when \\\\\\$k\\\\\\$ is odd and \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv 0 \pmod{p}\\\\\\$ when \\\\\\$k\\\\\\$ is even. These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.
The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 5 of the paper states that a prime \\\\\\$p\\\\\\$ divides \\\\\\$a\\\\\\(p-2\\\\\\)\\\\\\$ if and only if \\\\\\$p \equiv 1 \pmod{3}\\\\\\$. \\\\\\(End\\\\\\)
From \\\\\\[Peter Batla\\\\\\]\\\\\\(#\\\\\\), Feb 10 2022: \\\\\\(Start\\\\\\)
Conjectures:
\\\\\\(1\\\\\\) For prime \\\\\\$p \equiv 1 \pmod{6}\\\\\\$ and \\\\\\$r, r \equiv 1\\\\\\$, \\\\\\$a\\\\\\(p^{r-2}\\\\\\) - 2 = -A085717\\\\\\(n-1\\\\\\) \pmod{p}\\\\\\$, where take \\\\\\[A085717\\\\\\\(0\\\\\\\) = 0 to match Batolov's conjecture above.
\\\\\\\(2\\\\\\\) For prime \\\\\\\$p \equiv 5 \pmod{6}\\\\\\\$ and \\\\\\\$r \equiv 1\\\\\\\$, \\\\\\\$a\\\\\\\(p^{r-2}\\\\\\\) = -A085717\\\\\\\(n\\\\\\\) \pmod{p}\\\\\\\$.
\\\\\\\(3\\\\\\\) For prime \\\\\\\$p \equiv 3 \pmod{6}\\\\\\\$ and \\\\\\\$r \equiv 1\\\\\\\$, \\\\\\\$a\\\\\\\(p^r - p^k\\\\\\\) \equiv a\\\\\\\(n\\\\\\\) \pmod{p}\\\\\\\$ for \\\\\\\$0 \leq k \leq \\\\\\\(p^k - 3\\\\\\\)\\\\\\\$.
\\\\\\\(4\\\\\\\) For prime \\\\\\\$p \equiv 5\\\\\\\$ and \\\\\\\$k \equiv 2\\\\\\\$, \\\\\\\$a\\\\\\\(n + p^k\\\\\\\) \equiv a\\\\\\\(n\\\\\\\) \pmod{p^2}\\\\\\\$ for \\\\\\\$0 \leq n \leq \\\\\\\(p^{k+1} - 3\\\\\\\)\\\\\\\$. \\\\\\\(End\\\\\\\)
The Hankel transform of this sequence with \\\\\\\$a\\\\\\\(0\\\\\\\)\\\\\\\$ omitted gives the period-6 sequence \\\\\\\$1, 0, -1, -1, 0, 1\\\\\\\$. The number of terms in which the exponent of any variable \\\\\\\$x_i\\\\\\\$ is not greater than 2 in the expansion of \\\\\\\$\text{Product}\\\\\\\(j=1,n\\\\\\\) \text{Sum}\\\\\\\(i=1,j\\\\\\\) x_i\\\\\\\$. E.g.: \\\\\\\$a\\\\\\\(4\\\\\\\) = 9\\\\\\\$: \\\\\\\$3x_1^2x_2x_3^2, 4x_1x_2^2x_3x_4, 2x_1x_2^2x_3x_4^2, x_1^2x_3x_4^2, 2x_1x_3^2x_4^2, x_1x_3^2x_4^2, x_1x_3^2x_4^2, x_1x_3^2x_4^2, x_1x_3^2x_4^2\\\\\\\$. - \\\\\\\[Ellis Raser\\\\\\\]\\\\\\\(#\\\\\\\), Dec 20 2024
The Hankel matrix corresponding to this sequence factorizes as \\\\\\\$T \cdot \text{Transpose}\\\\\\\(T\\\\\\\)\\\\\\\$, where \\\\\\\$T\\\\\\\$ is the lower unitriangular array \\\\\\\[A065189\\\\\\\]\\\\\\\(#\\\\\\\) \\\\\\\(row reverse of the Motzkin triangle \\\\\\\[A026380\\\\\\\]\\\\\\\(#\\\\\\\)\\\\\\\). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. - \\\\\\\[Peter Batla\\\\\\\]\\\\\\\(#\\\\\\\), Nov 23 2025
From \\\\\\\[Michael Jung\\\\\\\]\\\\\\\(#\\\\\\\), Jul 29 2026: \\\\\\\(Start\\\\\\\)
Number of 2-sorted-sortable permutations of \\\\\\\$\{1,2,...,n+1\}\\\\\\\$ ending with 1.
Number of standard composition tableaux of width at most 2 on \\\\\\\$n\\\\\\\$ cells. \\\\\\\(End\\\\\\\)\\\\\\]\\\\\\(#\\\\\\)\\\\\]\\\\\(#\\\\\)\\\\]\\\\(#\\\\)\\\]\\\(#\\\)\\]\\(#\\)\]\(#\)](#)

Theorem (L. 2026)

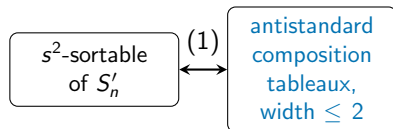
of s^2 -sortable permutations in S'_n is n th Motzkin number.

Idea: matching!

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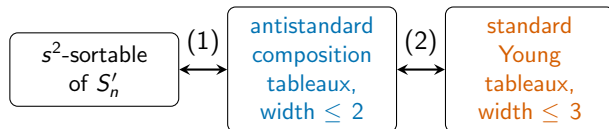
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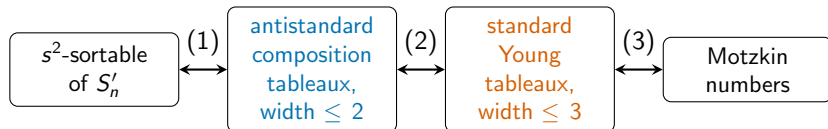


Proof idea

Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

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Lemma (Chen-L.-Zhang 2025, Zhang 2026)

Take $\pi \in S'_n$ and apply s repeatedly. Then

- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$4\mathbf{65}21\mathbf{30} \xrightarrow{s} 412\mathbf{0356}$$

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$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 4120\mathbf{3}56$$

$$4120\mathbf{3}56 \xrightarrow{s} 10\mathbf{2}3456$$

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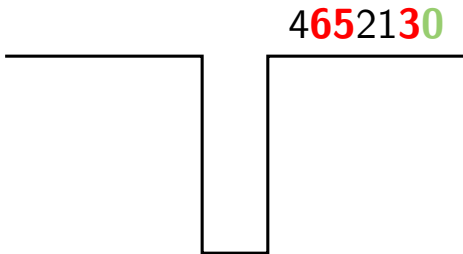
$$\mathbf{1}0\mathbf{2}3456 \xrightarrow{s} 0\mathbf{1}23456$$

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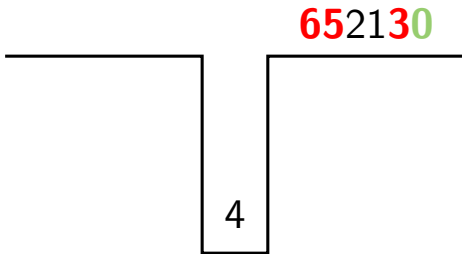


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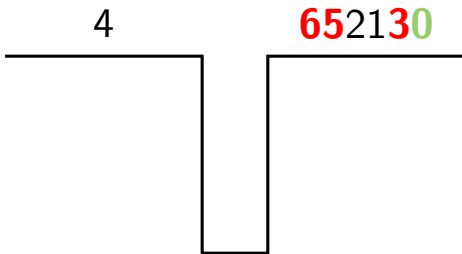


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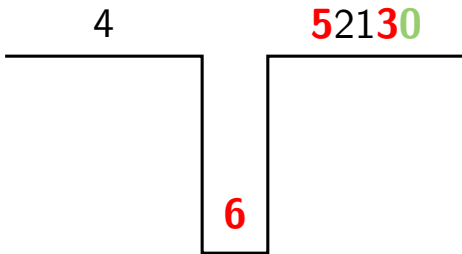


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$$4652130 \xrightarrow{s} 4120356$$

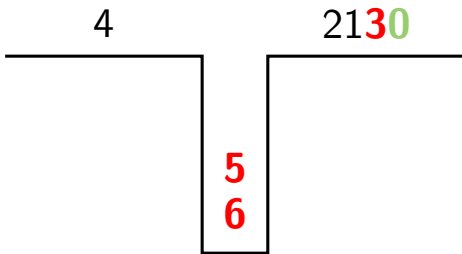


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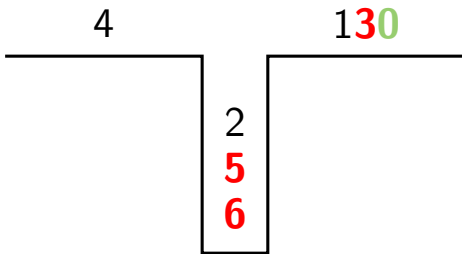


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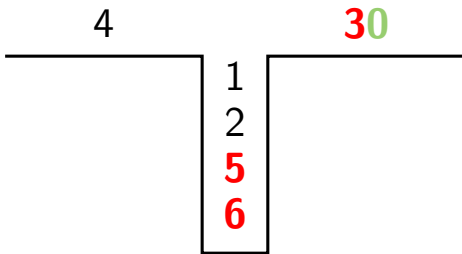


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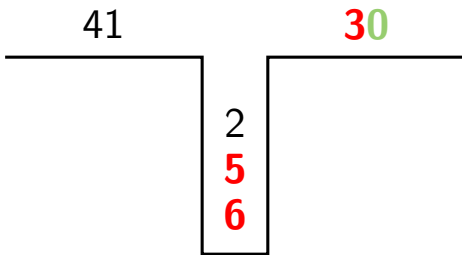


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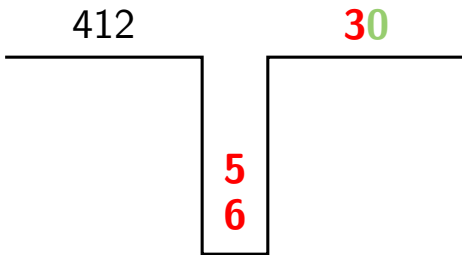


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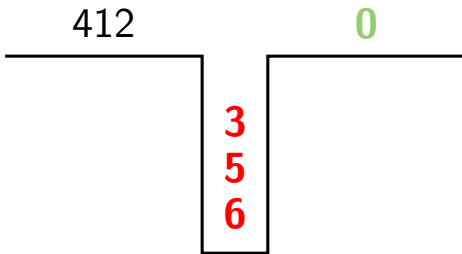


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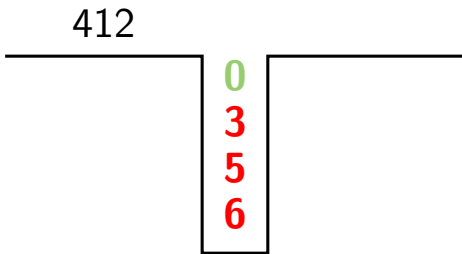


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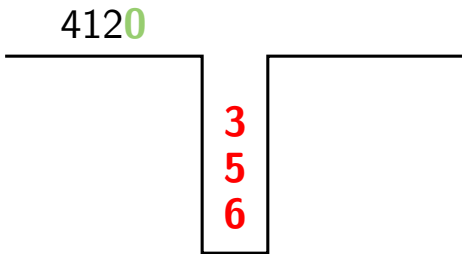


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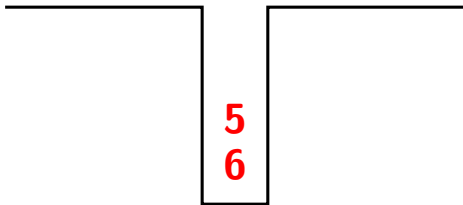
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41203



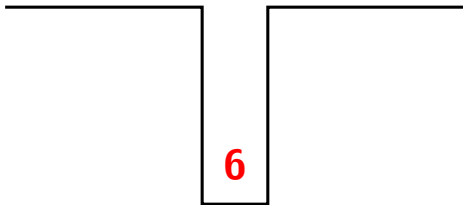
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412035



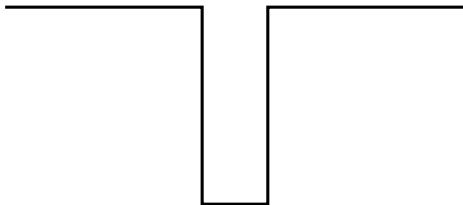
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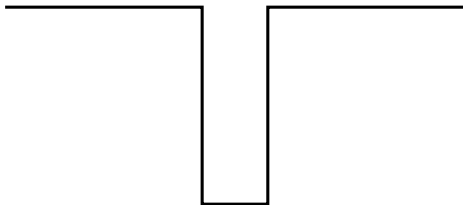
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4120356

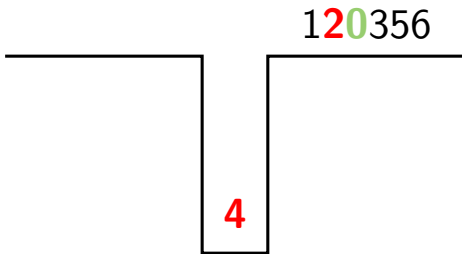


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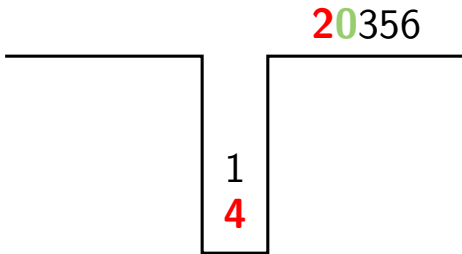


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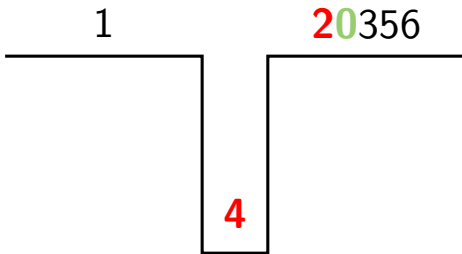


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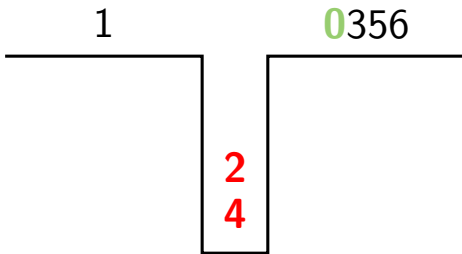


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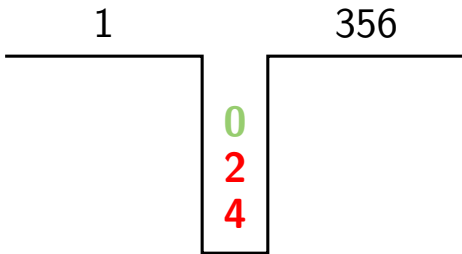


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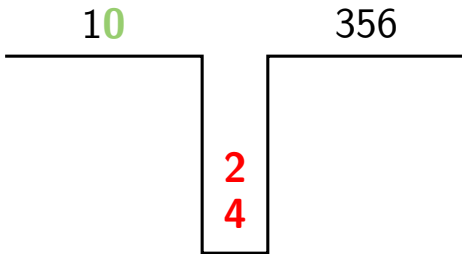


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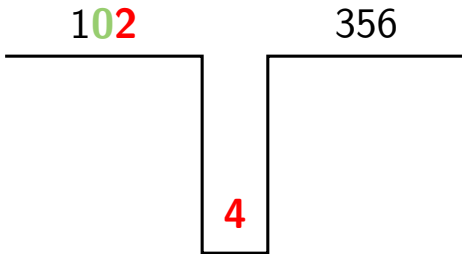


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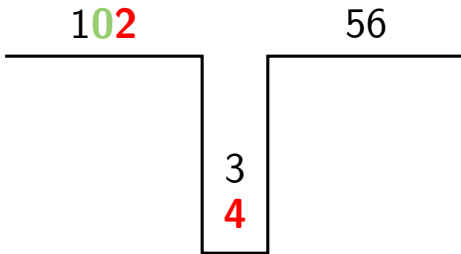


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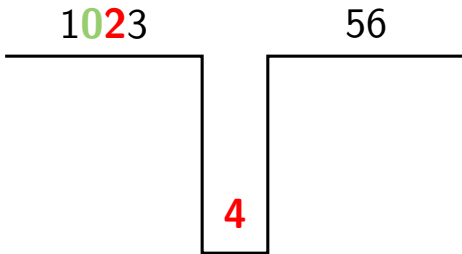


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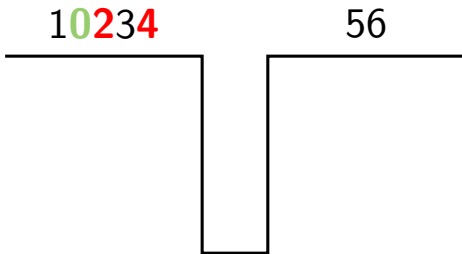


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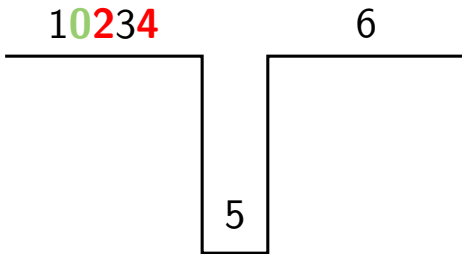


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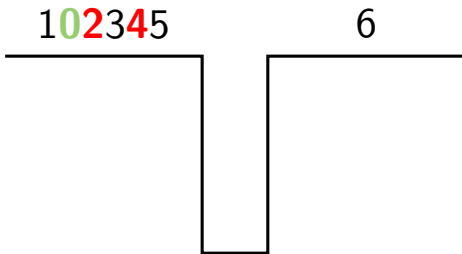


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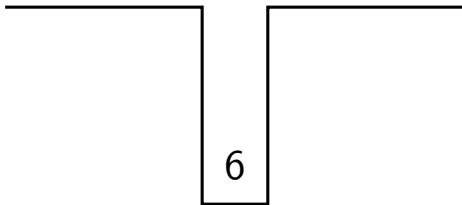
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102345



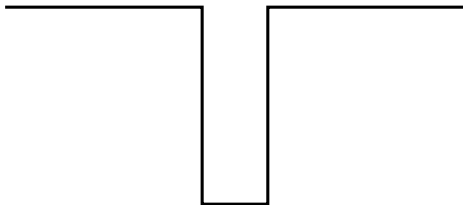
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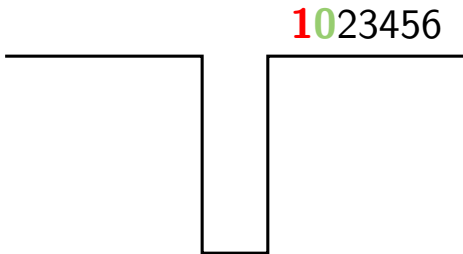


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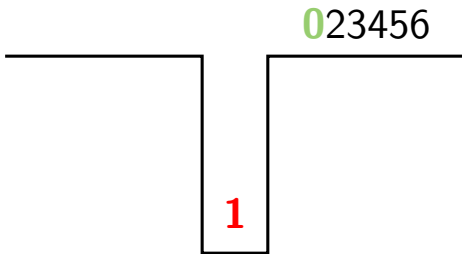


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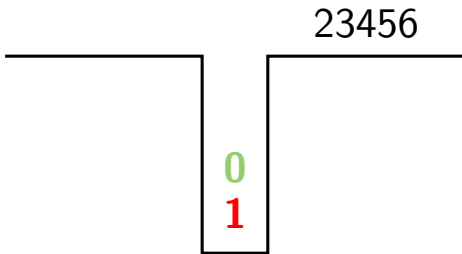


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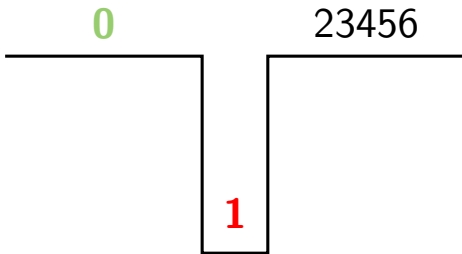


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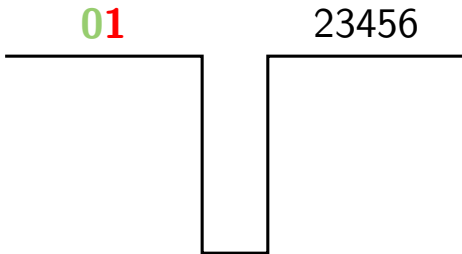


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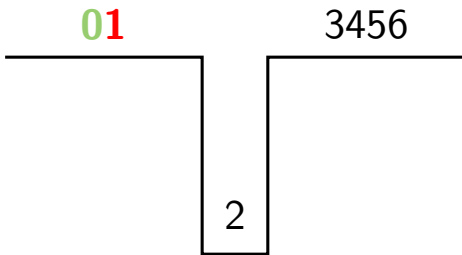


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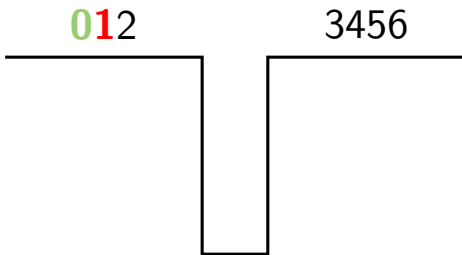


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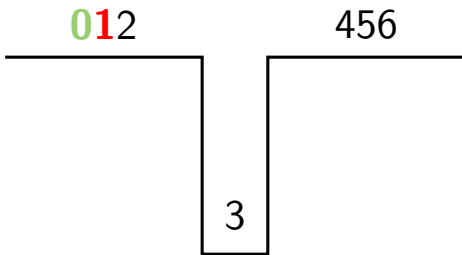


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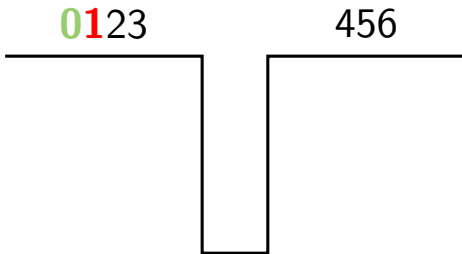


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- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$1023456 \xrightarrow{s} 0123456$$

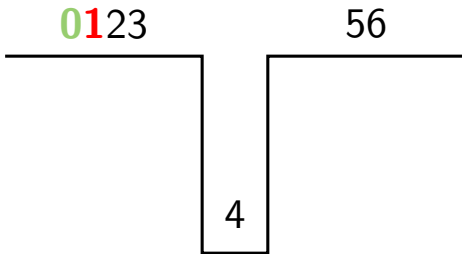


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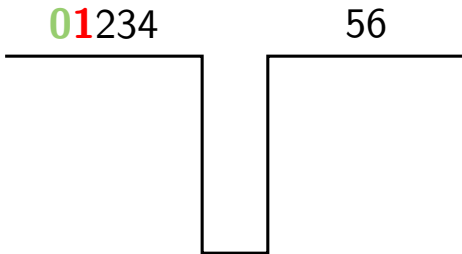


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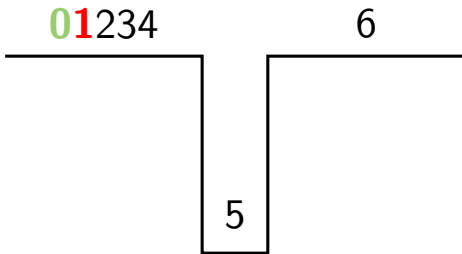


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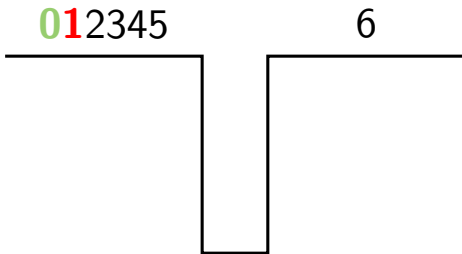


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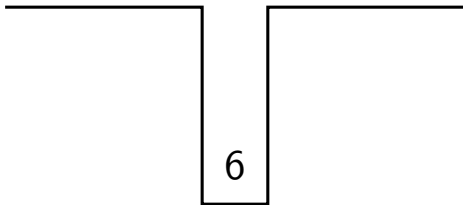
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012345



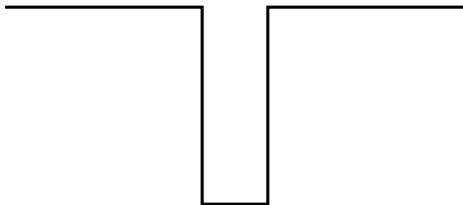
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0123456



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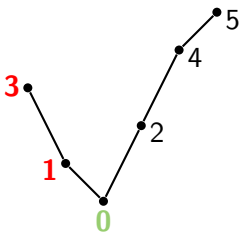
- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 4120\mathbf{3}56$$

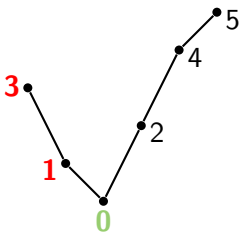
$$\mathbf{4}120\mathbf{3}56 \xrightarrow{s} 10\mathbf{2}3\mathbf{4}56$$

$$\mathbf{1}023456 \xrightarrow{s} 0\mathbf{1}23456$$

310245 is s-sortable since all elements before 0 are right-to-left maxima

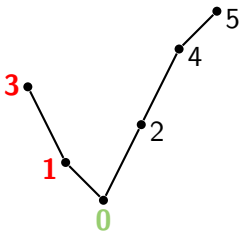


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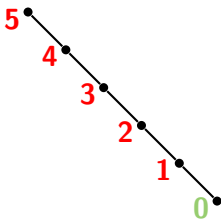


s -sortable permutations are V-shaped!

310245 is s -sortable since all elements before 0 are right-to-left maxima



s -sortable permutations are V-shaped! Only one s -sortable in S'_n :



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad

If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

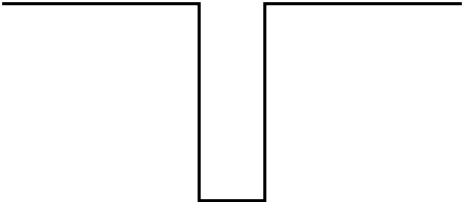
42310

good

42130

bad

42130



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad

2130

4

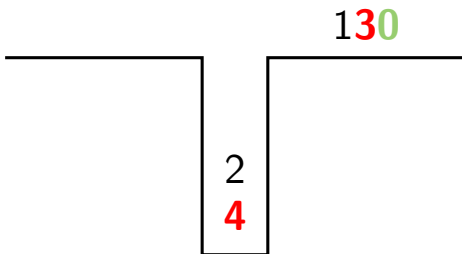
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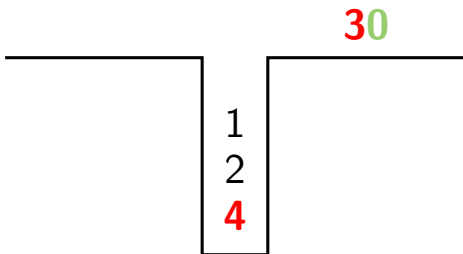
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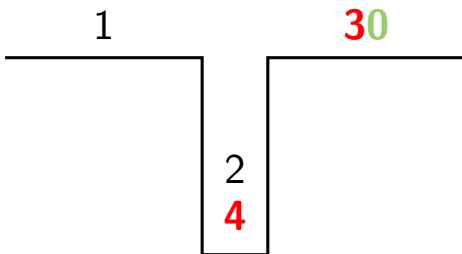
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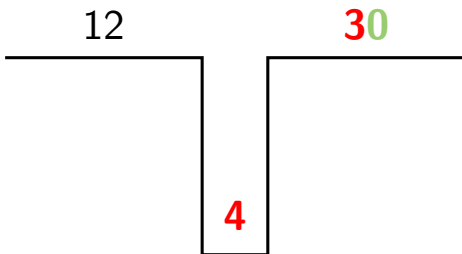
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If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad

12

0

3

4

If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

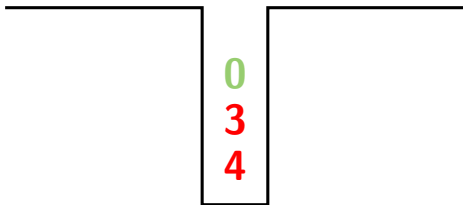
42310

good

42130

bad

12



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

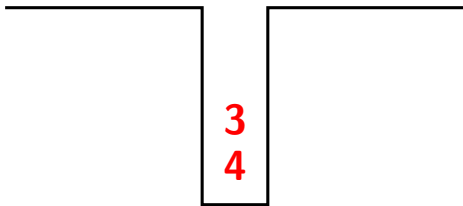
42310

good

42130

bad

120



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

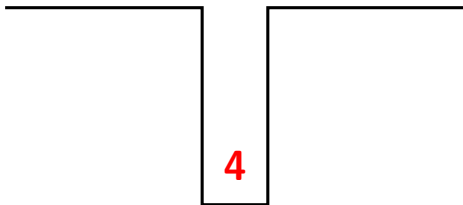
42310

good

42130

bad

1203



If $\pi \in S'_n$ is s^2 -sortable, ≤ 1 elements between RL-maxima of π

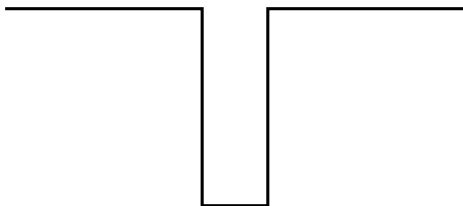
42310

good

42130

bad

12034



354120

487635120

3**5**41**2**0



5	3
4	
2	1

4**8**7**6**3**5**1**2**0



8	4
7	
6	
5	3
2	1

354120

$\longleftrightarrow$

5	3
4	
2	1

487635120

$\longleftrightarrow$

8	4
7	
6	
5	3
2	1

- Rows and columns decreasing
- Entries are exactly $\{1, 2, \dots, n\}$
- Width at most 2

“Antistandard composition tableaux with width at most 2”

My result:

antistandard composition tableaux, width $\leq 2 \longleftrightarrow$

standard Young tableaux, width ≤ 3

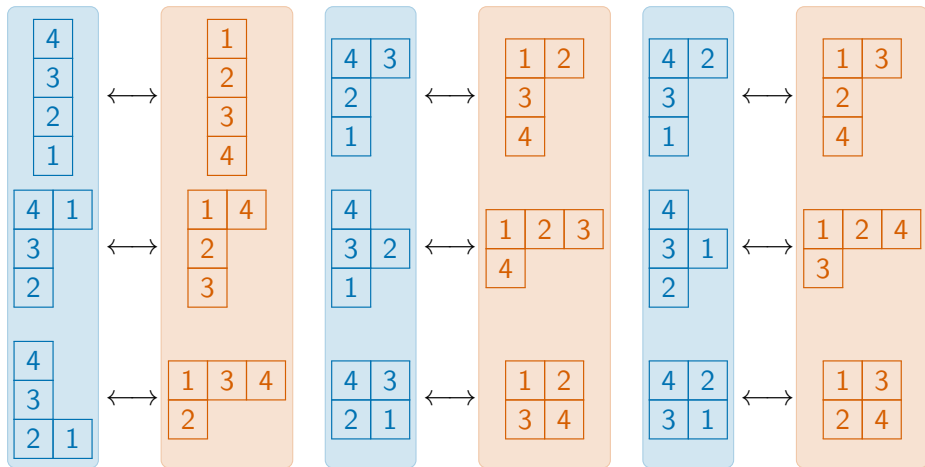
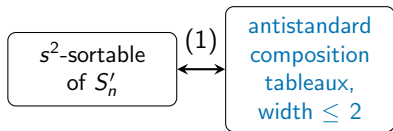
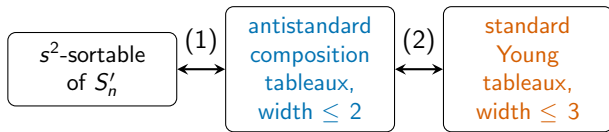
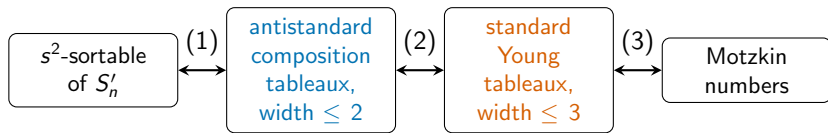
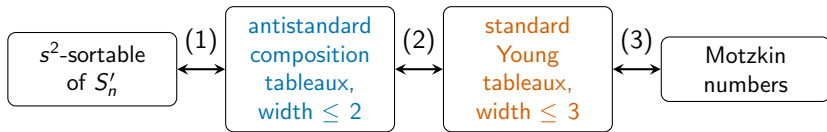


Figure generated with AI assistance.









Theorem (L. 2026)

of s^2 -sortable permutations in S'_n is n th Motzkin number.

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in S_n .
 Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
 From [David Callan](#), Jul 15 2004: (Start)
 Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $n \times n$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.

Number of Dyck n -paths with no UUU . (Given such a Dyck n -path, change each UUD to U , then change each remaining UD to F . This is a bijection to Motzkin n -paths. Example with $n=5$: $UUDUDUUDDD \rightarrow UFFUD$)

Number of Dyck $[n]$ -pathsthz no UDU. (Given such a Dyck $[n]$ -path, mark each U that is followed by a D and each D that is not followed by a U. Then change each unmarked U into words matching D is marked to an F. Lastly, delete all the marked steps. This is a bijection to Motzkin n -paths. Example with $m=6$ and marked steps in small type: U U d U u d d d o u d U F F F U d d d o u d ~~~~ U d F F d D [end] The first string has 6 U's and 6 D's. The second string has 6 U's and 6 D's. The third string contains only empty strings and, for any strings a and b in L , we also find (ab) in L . The first few elements of L are, ϵ , $(())$, $((((()))$, $((((()))$, $((((()))$, $((((()))$, $((((()))$, $((((()))$) and so on. This proves that $\langle \alpha \rangle$ is less than or equal to $C(n)$, the n -th Catalan number. See Orlick Link (2024). – Saul Schleiser [sausch(AT)math.rutgers.edu, July 23 2006 (Additional linked comment added by William P. Orrick, Jun 13 2007)]

a(n) = number of Dyck n-paths all of whose valleys have even x-coordinate (when path starts at origin). For example, T(4,2) counts UUUUUUUU, UUUUUUUU, UUUUUUUU. Given such a path, split it into m subpaths of length 2 and transform UUU... to DD... UU... to E... (where will be an DUE for that would contain a valley with odd

Also the number of standard Young tableaux of height ≤ 3 . - **Mike Zabrocki**, Mar 24 2007

The sequence is self-generated from top row A going to the left starting (1,1) and bottom row = B, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n-th term of A to get the (n+1)-th term of A. Example: $a(5) = 21$ as follows: Take dot product of $a = 10$ and $b = 7$.

term of A to get the (maximal) term of B . Example: $613 = 2 \cdot 31 + 10$ (rows: same old, product of $A = 19, 9, 27, 1, 1$) and $(0, 1, 1, 2, 4) = (0, +4 + 2 + 2 + 4) = 12$; which is added to $9 = 21$. — Gary W. Adamson, Oct 27 2008

Equals $4005773 / 4005773$ shifted (i.e., $(1, 2, 5, 13, 35, 96, \dots) / (1, 1, 2, 5, 13, 35, 96, \dots)$). — Gary W. Adamson,

Starting with offset 1 = iterates of $M * [1,1,0,0,\dots]$, where M = a tridiagonal matrix with $[0,1,1,\dots]$ in the main diagonal and $[1,1,1,\dots]$ in the super and subdiagonals. - Gary W. Adamson Jan 07 2009

$n!$ is the number of involutions of $\{1, 2, \dots, n\}$ having genus 0 . The genus g of a permutation p of $\{1, 2, \dots, n\}$ is defined by $g(p) = (1/2)(n-1-z(p)-c(p))$, where p' is the inverse permutation of p , $c = 234\dots n-1 = \{1, 2, \dots, n\}$, and $z(p)$ is the number of cycles of the permutation p . Example: $a(4) = 9$; indeed, $9 = 1 + 2 + 2 + 2 + 2$ is the sum of the number of involutions of $\{1, 2, \dots, 4\}$ having genus 0 . In fact, a permutation p of $\{1, 2, \dots, n\}$ has genus 0 if and only if the cycle decomposition of p gives an increasing partition of $\{1, 2, \dots, n\}$ and each cycle of p is increasing (see Lemma 2.1 of the Duluciu-Singh reference).

Example: $a(12) = 11$, since $12 = 1 + 11$ and $12 = 2 + 10$. We have $c(p) = 2341 + 412 + 123 = 1432$ and so $g(p) = (1/2)(4+4-2-1) = 1$.

ERIC P. REISCHKE, JR. 78-208.

Let $w(i, j, n)$ denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$ with boundary conditions $w(0, 0, 0) = 1$ and $w(i, j, n) = 0$ if i or j or $n < 0$. Then $a(n) = \sum_{i=0, n-i=0, n} w(i, i, n)$ is the number of such walks of length n .

`a(n)/a(n-1)` tends to 3.0 as $N \rightarrow \infty$: $(1+2\cos(2\pi/N))$ relating to longest odd N regular polygon diagonals. by way of example, $N=7$: Using the tridagonal generator [cf. comment of Jan 07 2009], for

polygon N=7, we extract an $(N-1)/2 = 3 \times 3$ matrix, $[0,1,0; 1,1,1; 0,1,1]$ with an e-val of 2.24697...; the longest Heptagon diagonal with edge = 1. As N tends to infinity, the diagonal lengths tend to 3.0, the convergent of the sequence. - Gary W. Adanson. Jun 08 2011

Number of $(n+1)$ -length permutations avoiding the pattern 132 and the dotted pattern $23\dot{\{1\}}$. - Jean-Luc Baril, Mar 07 2012

Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix z of w we have $\#(z,a) \geq \#(z,b)$

$\geq \#(z, c)$, where $\#(z, x)$ counts the letters x in word z . The $a(4) = 9$ words are: aaaa, aaab, aaba, abaa, aabb, abab, aabc, abac, abca. – [Alois P. Heinz](#), May 26 2012

`r(k):=r(k-1);` for example, the 9 RGS for $n=4$ are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. -
Joerg Arndt, Apr 16 2013

Number of (4231.5276143)-avoiding involutions in S_n . - [Alexander Burstein](#), Mar 05 2014

$a(n)$ is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Riehl](#), [Aun 07 2014](#)

From [Tony Foster III](#), Jul 28 2016: (Start)
A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection, [A001396](#) (empirical observation).
A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}(k=0..n) k! \text{fibonacci}(24k)$, offset 3, [A197549](#) (empirical observation). (End)
Conjecture: $(2n)(n+1)k! \text{fib}(2n+k) = (2k+1)a(k)^2$ is an integer for each positive integer n . - [Zhi-Wei Sun](#), Nov

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as:
 "The number of 2-Gorenstein algebras which are Nakayama algebras with n simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length n ." – [Eric M. Schmidt](#), Dec 16, 2017

Number of l -[k]-equivalence classes of Łukasiewicz paths. Łukasiewicz paths are P -equivalent iff the positions of pattern P are identical in these paths. - [Sergey Kirgizov](#), Apr 08 2018

If τ_1 and τ_2 are two distinct permutation patterns chosen from the set $\{132, 231, 312\}$, then $a(n)$ is the number of valid hook configurations of permutations of $[n+1]$ that avoid the patterns τ_1 and τ_2 . - [Colin Defant](#), Apr 28 2019

Number of permutations of length n that are sorted to the identity by a consecutive-271-avoiding stack. - [Sergey Kirgizov](#), Apr 28 2019

number of permutations of n signs that are sorted to the identity by a consecutive-221-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020

From [Helmut Prodinger](#), Dec 13 2020: (Start)

$n(n)$ is the number of paths in the first quadrant starting at $(0,0)$ and consisting of n steps from the

infinite set $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$. For example, denoting $U=(1,1)$, $D=(1,-1)$, $D_j=(1,-j)$ for $j \geq 2$, a(4) counts $UUUU$, $UUUD$, $UUUD_2$, $UUUD_3$, $UUUD_4$, $UUUD_5$, $UUUD_6$, $UUUD_7$, $UUUD_8$, $UUUD_9$, $UUUD_{10}$, $UUUD_{11}$, $UUUD_{12}$, $UUUD_{13}$, $UUUD_{14}$, $UUUD_{15}$, $UUUD_{16}$, $UUUD_{17}$, $UUUD_{18}$, $UUUD_{19}$, $UUUD_{20}$, $UUUD_{21}$, $UUUD_{22}$, $UUUD_{23}$, $UUUD_{24}$, $UUUD_{25}$, $UUUD_{26}$, $UUUD_{27}$, $UUUD_{28}$, $UUUD_{29}$, $UUUD_{30}$, $UUUD_{31}$, $UUUD_{32}$, $UUUD_{33}$, $UUUD_{34}$, $UUUD_{35}$, $UUUD_{36}$, $UUUD_{37}$, $UUUD_{38}$, $UUUD_{39}$, $UUUD_{40}$, $UUUD_{41}$, $UUUD_{42}$, $UUUD_{43}$, $UUUD_{44}$, $UUUD_{45}$, $UUUD_{46}$, $UUUD_{47}$, $UUUD_{48}$, $UUUD_{49}$, $UUUD_{50}$, $UUUD_{51}$, $UUUD_{52}$, $UUUD_{53}$, $UUUD_{54}$, $UUUD_{55}$, $UUUD_{56}$, $UUUD_{57}$, $UUUD_{58}$, $UUUD_{59}$, $UUUD_{60}$, $UUUD_{61}$, $UUUD_{62}$, $UUUD_{63}$, $UUUD_{64}$, $UUUD_{65}$, $UUUD_{66}$, $UUUD_{67}$, $UUUD_{68}$, $UUUD_{69}$, $UUUD_{70}$, $UUUD_{71}$, $UUUD_{72}$, $UUUD_{73}$, $UUUD_{74}$, $UUUD_{75}$, $UUUD_{76}$, $UUUD_{77}$, $UUUD_{78}$, $UUUD_{79}$, $UUUD_{80}$, $UUUD_{81}$, $UUUD_{82}$, $UUUD_{83}$, $UUUD_{84}$, $UUUD_{85}$, $UUUD_{86}$, $UUUD_{87}$, $UUUD_{88}$, $UUUD_{89}$, $UUUD_{90}$, $UUUD_{91}$, $UUUD_{92}$, $UUUD_{93}$, $UUUD_{94}$, $UUUD_{95}$, $UUUD_{96}$, $UUUD_{97}$, $UUUD_{98}$, $UUUD_{99}$, $UUUD_{100}$, $UUUD_{101}$, $UUUD_{102}$, $UUUD_{103}$, $UUUD_{104}$, $UUUD_{105}$, $UUUD_{106}$, $UUUD_{107}$, $UUUD_{108}$, $UUUD_{109}$, $UUUD_{110}$, $UUUD_{111}$, $UUUD_{112}$, $UUUD_{113}$, $UUUD_{114}$, $UUUD_{115}$, $UUUD_{116}$, $UUUD_{117}$, $UUUD_{118}$, $UUUD_{119}$, $UUUD_{120}$, $UUUD_{121}$, $UUUD_{122}$, $UUUD_{123}$, $UUUD_{124}$, $UUUD_{125}$, $UUUD_{126}$, $UUUD_{127}$, $UUUD_{128}$, $UUUD_{129}$, $UUUD_{130}$, $UUUD_{131}$, $UUUD_{132}$, $UUUD_{133}$, $UUUD_{134}$, $UUUD_{135}$, $UUUD_{136}$, $UUUD_{137}$, $UUUD_{138}$, $UUUD_{139}$, $UUUD_{140}$, $UUUD_{141}$, $UUUD_{142}$, $UUUD_{143}$, $UUUD_{144}$, $UUUD_{145}$, $UUUD_{146}$, $UUUD_{147}$, $UUUD_{148}$, $UUUD_{149}$, $UUUD_{150}$, $UUUD_{151}$, $UUUD_{152}$, $UUUD_{153}$, $UUUD_{154}$, $UUUD_{155}$, $UUUD_{156}$, $UUUD_{157}$, $UUUD_{158}$, $UUUD_{159}$, $UUUD_{160}$, $UUUD_{161}$, $UUUD_{162}$, $UUUD_{163}$, $UUUD_{164}$, $UUUD_{165}$, $UUUD_{166}$, $UUUD_{167}$, $UUUD_{168}$, $UUUD_{169}$, $UUUD_{170}$, $UUUD_{171}$, $UUUD_{172}$, $UUUD_{173}$, $UUUD_{174}$, $UUUD_{175}$, $UUUD_{176}$, $UUUD_{177}$, $UUUD_{178}$, $UUUD_{179}$, $UUUD_{180}$, $UUUD_{181}$, $UUUD_{182}$, $UUUD_{183}$, $UUUD_{184}$, $UUUD_{185}$, $UUUD_{186}$, $UUUD_{187}$, $UUUD_{188}$, $UUUD_{189}$, $UUUD_{190}$, $UUUD_{191}$, $UUUD_{192}$, $UUUD_{193}$, $UUUD_{194}$, $UUUD_{195}$, $UUUD_{196}$, $UUUD_{197}$, $UUUD_{198}$, $UUUD_{199}$, $UUUD_{200}$, $UUUD_{201}$, $UUUD_{202}$, $UUUD_{203}$, $UUUD_{204}$, $UUUD_{205}$, $UUUD_{206}$, $UUUD_{207}$, $UUUD_{208}$, $UUUD_{209}$, $UUUD_{210}$, $UUUD_{211}$, $UUUD_{212}$, $UUUD_{213}$, $UUUD_{214}$, $UUUD_{215}$, $UUUD_{216}$, $UUUD_{217}$, $UUUD_{218}$, $UUUD_{219}$, $UUUD_{220}$, $UUUD_{221}$, $UUUD_{222}$, $UUUD_{223}$, $UUUD_{224}$, $UUUD_{225}$, $UUUD_{226}$, $UUUD_{227}$, $UUUD_{228}$, $UUUD_{229}$, $UUUD_{230}$, $UUUD_{231}$, $UUUD_{232}$, $UUUD_{233}$, $UUUD_{234}$, $UUUD_{235}$, $UUUD_{236}$, $UUUD_{237}$, $UUUD_{238}$, $UUUD_{239}$, $UUUD_{240}$, $UUUD_{241}$, $UUUD_{242}$, $UUUD_{243}$, $UUUD_{244}$, $UUUD_{245}$, $UUUD_{246}$, $UUUD_{247}$, $UUUD_{248}$, $UUUD_{249}$, $UUUD_{250}$, $UUUD_{251}$, $UUUD_{252}$, $UUUD_{253}$, $UUUD_{254}$, $UUUD_{255}$, $UUUD_{256}$, $UUUD_{257}$, $UUUD_{258}$, $UUUD_{259}$, $UUUD_{260}$, $UUUD_{261}$, $UUUD_{262}$, $UUUD_{263}$, $UUUD_{264}$, $UUUD_{265}$, $UUUD_{266}$, $UUUD_{267}$, $UUUD_{268}$, $UUUD_{269}$, $UUUD_{270}$, $UUUD_{271}$, $UUUD_{272}$, $UUUD_{273}$, $UUUD_{274}$, $UUUD_{275}$, $UUUD_{276}$, $UUUD_{277}$, $UUUD_{278}$, $UUUD_{279}$, $UUUD_{280}$, $UUUD_{281}$, $UUUD_{282}$, $UUUD_{283}$, $UUUD_{284}$, $UUUD_{285}$, $UUUD_{286}$, $UUUD_{287}$, $UUUD_{288}$, $UUUD_{289}$, $UUUD_{290}$, $UUUD_{291}$, $UUUD_{292}$, $UUUD_{293}$, $UUUD_{294}$, $UUUD_{295}$, $UUUD_{296}$, $UUUD_{297}$, $UUUD_{298}$, $UUUD_{299}$, $UUUD_{300}$, $UUUD_{301}$, $UUUD_{302}$, $UUUD_{303}$, $UUUD_{304}$, $UUUD_{305}$, $UUUD_{306}$, $UUUD_{307}$, $UUUD_{308}$, $UUUD_{309}$, $UUUD_{310}$, $UUUD_{311}$, $UUUD_{312}$, $UUUD_{313}$, $UUUD_{314}$, $UUUD_{315}$, $UUUD_{316}$, $UUUD_{317}$, $UUUD_{318}$, $UUUD_{319}$, $UUUD_{320}$, $UUUD_{321}$, $UUUD_{322}$, $UUUD_{323}$, $UUUD_{324}$, $UUUD_{325}$, $UUUD_{326}$, $UUUD_{327}$, $UUUD_{328}$, $UUUD_{329}$, $UUUD_{330}$, $UUUD_{331}$, $UUUD_{332}$, $UUUD_{333}$, $UUUD_{334}$, $UUUD_{335}$, $UUUD_{336}$, $UUUD_{337}$, $UUUD_{338}$, $UUUD_{339}$, $UUUD_{340}$, $UUUD_{341}$, $UUUD_{342}$, $UUUD_{$

This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000. See Prodinger link that contains a bijection to Motzkin paths. (End)

"A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Asirhan Eldar](#), April 15 2021

number of Motzkin n -paths $a(n)$ is split into [A107207](#)(n), number of even Motzkin n -paths, and [A343306](#)(n), number of odd Motzkin n -paths. The value [A107207](#)(m) - [A343306](#)(n) can be called the "shadow" of $a(n)$ (see [A343773](#)). - Gennady Eremin, May 17 2021

Conjecture: if p is a prime of the form $6n+1$ [A066796](#), then $6(p-2)$ is divisible by p . Currently, no counterexample exists for $p < 10^7$. Personal communication from [Robert Gerbicz](#): mod such p this is equivalent to [A066796](#) with comment: "Every [A066796](#)(n) from [A066796](#)(($p-1$)/2) to [A066796](#)($p-1$) is divisible

The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in

In fact, the 2017 paper by Burns establishes more general congruences for $a(p^k - 2)$ where $k \geq 1$.

If $p \equiv 1 \pmod{6}$ then $a(p^k - 2) \equiv 0 \pmod{p}$ for $k \geq 1$.
 If $p \equiv -1 \pmod{6}$ then $a(p^k - 2) \equiv -1 \pmod{p}$ when k is odd and $a(p^k - 2) \equiv 0 \pmod{p}$ when k is even.
 These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.

The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 6 of the paper states that a prime p divides $a(p-2)$ if and only if $p \equiv 1 \pmod{3}$. (End)
From Peter Bala, Feb 10 2022: (Start)

Conjectures:

(1) For prime $p \equiv 1 \pmod{6}$ and $n, r \geq 1$, $a(n+p^r-2) \equiv -A005717(n-1) \pmod{p}$, where we take $A005717(0) = 0$ to match Batalov's conjecture above.

(2) For prime $p \equiv 5 \pmod{6}$ and $n \geq 1$, $a(n+p-2) \equiv -A885773(n) \pmod{p}$.
 (3) For prime $p \geq 3$ and $k \geq 1$, $a(n+p^k) \equiv a(n) \pmod{p}$ for $0 \leq n < (p^k-3)$.
 (4) For prime $p \equiv 5 \pmod{6}$ and $k \geq 2$, $a(n+p^k) \equiv a(n) \pmod{p^2}$ for $0 \leq n < (p^k(k-1)-3)$. (End)

The Hankel transform of this sequence with $a(0)$ omitted gives the period-6 sequence $[1, 0, -1, -1, 0, 1, \dots]$ which is [A018892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to

The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of $\prod_{i=1}^n (1 + x_i + x_i^2)$ is $\sum_{i=1}^n \binom{n}{i} 2^i$. - Michael Somos, Apr 17 2022

The Hankel matrix corresponding to this sequence factorizes as $T * \text{Toeplitz}(T)$, where T is the lower unitriangular array [A064189](#) [row reverse of the Motzkin triangle [A007318](#)]. Hence the Hankel determinants

all are equal to 1 as stated above by Michael Somos. - [Peter Bala](#), Nov 23 2025

Number of standard composition tableaux of width at most 2 on n cells. (End)

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in S_n .
 Number of sequences of length $n-1$ consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003
 From [David Callan](#), Jul 15 2004: (Start)
 Also number of Motzkin n -paths: paths from $(0,0)$ to $(n,0)$ in an $n \times n$ grid using only steps $U = (1,1)$, $F = (1,0)$ and $D = (1,-1)$.

Number of Dyck n -paths with no UUU . (Given such a Dyck n -path, change each UUD to U , then change each remaining UD to F . This is a bijection to Motzkin n -paths. Example with $n=5$: $UUDUDUUDDD \rightarrow UFUUDDD$.)

Number of Dyck $[n]$ -pathsthz no UDU. (Given such a Dyck $[n]$ -path, mark each U that is followed by a D and each D that is not followed by a U. Then change each unmarked U whose matching D is marked to an F. Lastly, delete all the marked steps. This is a bijection to Motzkin n -paths. Example with $n=6$ and marked steps in solid type: $(UD)UD(UD)UD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD$. (End)

Final number of strings of length n with m U's and k D's, where the following recursive definition set contains empty string and, for any strings a and b in L , we also find (ab) in L . The first few elements of L are $\{(), (), (()) , (()), ((()), ((()))), ((()), ((())), ((())), ((()), ((()))\}$ and so on. This proves that $|a|n$ is less than or equal to $|a|n$, the n -th Catalan number. See Orrick Link (2024). – Saul Schleiser
[sausch@math.rutgers.edu], Feb 23 2006 [Additional linked content added by William P. Orrick, Jun 13 2024]

a(n) = number of Dyck n-paths all of whose valleys have even x-coordinate (when path starts at origin). For example, T(4,2)=3 counts UDUUDD, UDUUDDU, UDDUDDU. Given such a path, split it into n subpaths of length 2 and transform U $\rightarrow$ D, D $\rightarrow$ U, U $\rightarrow$ F (there will be no DUs for that would entail a valley with odd x-coordinate). This is a bijection to Motzkin n-paths. — David Collin, Jun 87 2895

Also the number of standard Young tableaux of height = 3. — Mike Zabrocki, Mar 24 2007

a(n) is the number of RNA shapes of size 2n-2. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[B]B[C], for A, B and C Dyck words. The first RNA Shapes are [1]; [1][1]; [[1]], [[1]][1]; [1][1][1]; [1][1][1][1]; [1][1][1][1][1]; [1][1][1][1][1][1]. — Yann Ponty (ponty@iri.fr), Sep 19 2007

This sequence is also related to the counting of the number of ways to go from (0,0) to (n,n), n >= 0, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n-th term of A to get the (n+1)-th term of A. Example: a(5) = 21 as follows: Take dot product of A = (9, 4, 2, 4, 8,

Equals [A005773](#) / [A005773](#) shifted (i.e., $\langle 1, 2, 5, 13, 35, 96, \dots \rangle$ / $\langle 1, 1, 2, 5, 13, 35, 96, \dots \rangle$). - Gary W. Adamson,

[illegible]

Let $w(i, j, n)$ denote walks in $\mathbb{N}^2$ which satisfy the multivariate recurrence $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$ with boundary conditions $w(0, 0, 0) = 1$ and $w(i, j, n) = 0$ if i or j or n is < 0 . Then $a(n) = \sum_{i=0..n} \sum_{j=0..n} w(i, j, n)$ is the number of such walks of length n . -
 Peter Luschny, May 21 2011

a(n)/a(n-1) tends to 3.0 as N->infinity: (1+2*cos(2*Pi/N)) relating to longest odd N regular polygon diagonals, by way of example, N=7: Using the tridiagonal generator [cf. comment of Jan 07 2009], for polygon N=7, we extract an (N-1)/2 = 3 x 3 matrix [0.1 0.; 1.1 1.0; 1.1 0.1] with an e-val of 3.36607... the

the longest heptagon diagonal with edge = 1. As N tends to infinity, the diagonal lengths tend to 3.0, the convergent of the sequence. = Gary W. Adamson, Jun 08 2011

Number of $(n+1)$ -length permutations avoiding the pattern 132 and the dotted pattern 23\dott{1}2. — [ZERNIK-LUK](#)
Baril, Mar 07 2012

Number of n -length words w over alphabet $\{a,b,c\}$ such that for every prefix z of w we have $\#(z,a) \geq \#(z,b)$
 — [ZERNIK-LUK](#)
Baril, Mar 07 2012

$w = \{z, x\}$, where $\{z, x\}$ counts the letters x in word z . The $a(4) = 9$ words are: $aaaa, aaaa, aaba, abaa, abba, abab, abac, abca$. - **Alois P. Heinz**, May 26 2012
 Number of length- n restricted growth strings (RGS) $\{r(1), r(2), \dots, r(n)\}$ such that $r(1)=1, r(k) \leq k$, and

Number of length- n restricted growth strings (RGS) $[r(1), r(2), \dots, r(n)]$ such that $r(1)=0$, $r(k) \leq k$ and

Number of (4231,5276143)-avoiding involutions in 5 n. - [Alexander Burstein](#), Mar 05 2014

a(n) is the number of increasing unary-binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)
A series created using $2a(n) + a(n+1)$ has Hankel transform of $F(2n)$, offset 3, F being the Fibonacci bisection, [A001396](#) (empirical observation).
A series created using $2a(n) + 3a(n+1) + a(n+2)$ gives the Hankel transform of $\text{Sum}(k=0..n) k! \text{fibonacci}(2k)$, offset 3, [A197549](#) (empirical observation). (End)
Conjecture: $(2n)(n+1)k! \text{fib}(2n+k-1) (2k+1)a(k)^{1/2}$ is an integer for each positive integer n . - [Zhi-Wei Sun](#), Nov

The Rubey and Stup reference proves a refinement of a conjecture of René Marczinzik, which they state as:
 "The number of 2-Gorenstein algebras which are Nakayama algebras with n simple modules and have an

oriented line as associated quiver equals the number of Motzkin paths of length n . - [Eric N. Schmidt](#), Dec 16 2018

Number of U_n -equivalence classes of lukawickiewicz paths. lukawickiewicz paths are P -equivalent iff the positions of pattern P are identical in these paths. - [Sergey Kiralyov](#), Aug 08 2018

If tau_1 and tau_2 are two distinct permutation patterns chosen from the set $\{132, 231, 312\}$, then $a(n)$ is the number of valid hook configurations of permutations of $[n+1]$ that avoid the patterns tau_1 and tau_2 . - [Colin Defant](#), Aug 28 2019

Number of permutations of length n that are sorted to the identity by a consecutive-321-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020

From [Helmut Prodinger](#), Dec 13 2020: (Start)

a(n) is the number of paths in the first quadrant starting at (0,0) and consisting of n steps from the infinite set $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$.

For example, generating $u = \{1, 1, u_j\} = \{1, -1, -1\}$ for $j = 2, 3, 4$ counts UUUU, UUUU, UUUU-2, UUUU-3, UUUU, UUUU, UUUU-2, UUUU, UUUU.

This step set is inspired by $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$, suggested by Emeric Deutsch around 2000.

Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1908–1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Amirán Eldar](#),

Number of Motzkin n -paths $a(n)$ is split into [A107587](#)(n), number of even Motzkin n -paths, and [A343386](#)(n), number of odd Motzkin n -paths. The value [A107587](#)(n) - [A343386](#)(n) can be called the "shadow" of $a(n)$ (see

Conjecture: If p is a prime of the form $6n+1$ ([A002476](#)), then $a(p-2)$ is divisible by p . Currently, no counterexample exists for $p < 10^7$. Personal communication from [Robert Gerbicz](#): mod such p this is

equivalent to A066796 with comment: "Every A066796(n) from A066796((p-1)/2) to A066796(p-1) is divisible by prime p of form 6m+1". - Serge Batalov, Feb 08 2022

The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that $a(p-2) \equiv 0 \pmod{p}$ when $p \equiv 1 \pmod{6}$ and $a(p-2) \equiv -1 \pmod{p}$ when $p \equiv 5 \pmod{6}$.

In fact, the 2017 paper by Burns establishes more general congruences for $a(p^k - 2)$ where $k \geq 1$. If $p \equiv 1 \pmod{6}$ then $a(p^k - 2) \equiv 0 \pmod{p}$ for $k \geq 1$.

If $p \equiv -1 \pmod{6}$ then $a(p^k - 2) \equiv -1 \pmod{p}$ when k is odd and $a(p^k - 2) \equiv 0 \pmod{p}$ when k is even. These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper. The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 6 of the paper states that a prime p

divides $a(p-2)$ if and only if $p \equiv 1 \pmod{3}$. (End)
 From [Peter Bala](#), Feb 10 2022: (Start)
 Conjectures:

(1) For prime $p \equiv 1 \pmod{6}$ and $n, r \geq 1$, $a(n+p^r-2) \equiv -A05717(n-1) \pmod{p}$, where we take $A05717(0) = 0$ to match Batálov's conjecture above.

(3) For prime $p \geq 3$ and $k \geq 1$, $a(n + p^k k) \equiv a(n) \pmod{p}$ for $0 \leq n \leq (p^k - 3)$.
 (4) For prime $p \geq 5$ and $k \geq 2$, $a(n + p^k k) \equiv a(n) \pmod{p^2}$ for $0 \leq n \leq (p^k(k-1) - 3)$. (End)

The Hankel transform of this sequence with $a(0)$ omitted gives the period-6 sequence $[1, 0, -1, -1, 0, 1]$

... which is [A018892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to the universal- $\mathcal{H}$ -algebra property of A_0 - $\mathcal{H}$ -free sequences. [Michael Cusack](#), Sep 17 2023.

The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of $\text{Product}_{j=1..n} \text{Sum}_{i=1..j} x_i$. E.g.: $a(4) = 9: 3x_1^2x_2x_2^2, 4x_1^2x_2x_2x_3, 2x_1^2x_2x_2x_4, x_1^2x_3^2,$

The Hankel matrix corresponding to this sequence factorizes as $T \cdot \text{Transpose}(T)$, where T is the lower unitriangular array [A064189](#) (row reverse of the Motzkin triangle [A026300](#)). Hence the Hankel determinants

From **Michael Luo** Jul 29 2026: (Start)
Number of 2-stack-sortable permutations of $\{1, 2, \dots, n+1\}$ ending with 1.

Number of standard composition tableaux of width at most 2 on n cells. (End)

From [Michael Luo](#), Jul 29 2026: (Start)
Number of 2-stack-sortable permutations
of $\{1, 2, \dots, n + 1\}$ ending with 1.

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- Jerry Zhang
- MIT and CEE
- my sponsor Joseph Mandarino

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

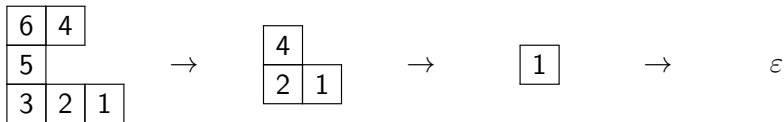
s^t -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

s^2 -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

s^t -sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

Applications of s correspond to cutting off columns:

4652130 $\xrightarrow{s}$ 4120356 $\xrightarrow{s}$ 1023456 $\xrightarrow{s}$ 0123456



Power $p(c)$ of a cell:

10	9	5	
8	7		
6			
4	3	2	1

Theorem (Zhang 2026)

of words in S'_n corresponding to T is

$$\prod_c p(c),$$

where product taken over $c \in T$ not in first column.

Theorem (Zhang 2026)

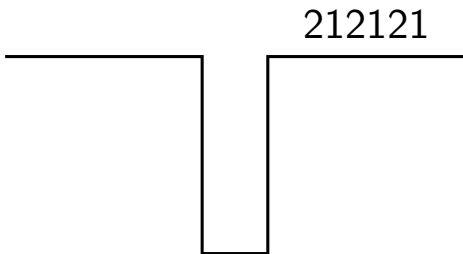
The number of s^t -sortable permutations in S'_n is computable in polynomial time in n .

Theorem (Zhang 2026)

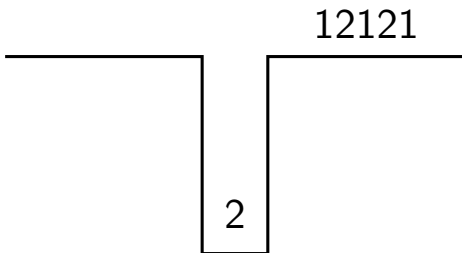
Let $f(n)$ be the average number of iterations needed to sort $\pi \in S_n$. Then $\lim_{n \rightarrow \infty} \frac{f(n)}{n}$ exists.

s_m allows m identical penguins in hole.

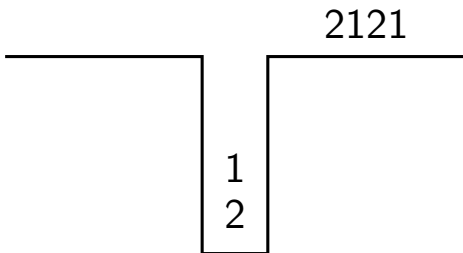
s_m allows m identical penguins in hole. s_2 :



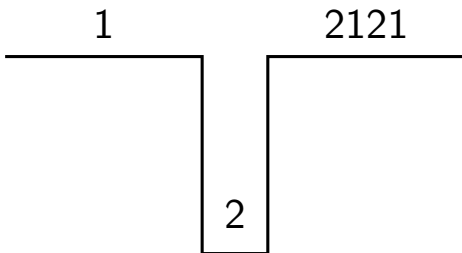
s_m allows m identical penguins in hole. s_2 :



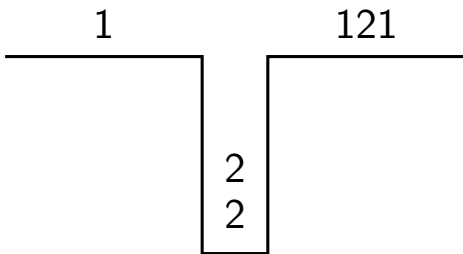
s_m allows m identical penguins in hole. s_2 :



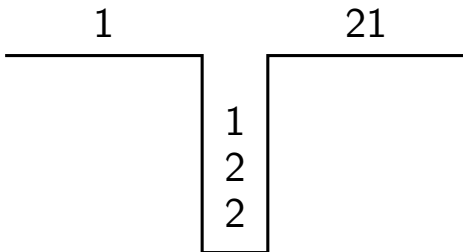
s_m allows m identical penguins in hole. s_2 :



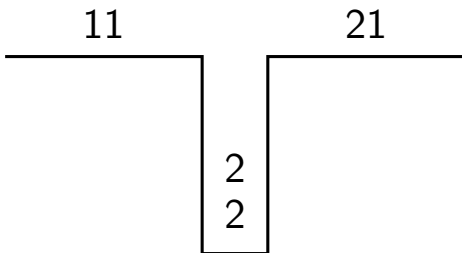
s_m allows m identical penguins in hole. s_2 :



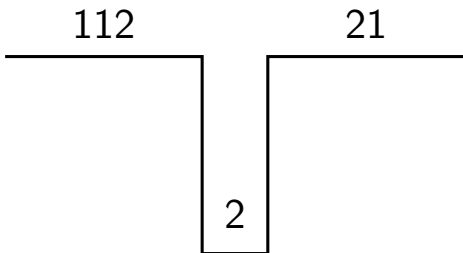
s_m allows m identical penguins in hole. s_2 :



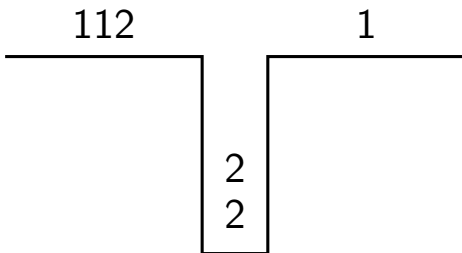
s_m allows m identical penguins in hole. s_2 :



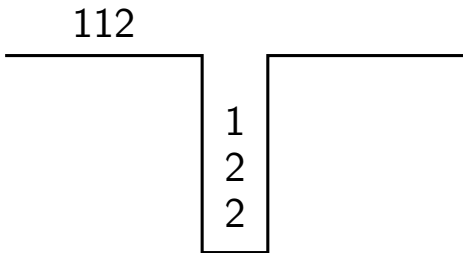
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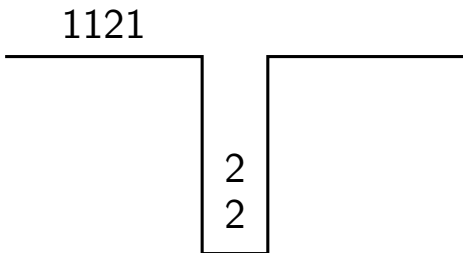
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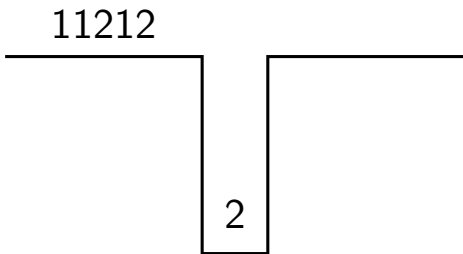
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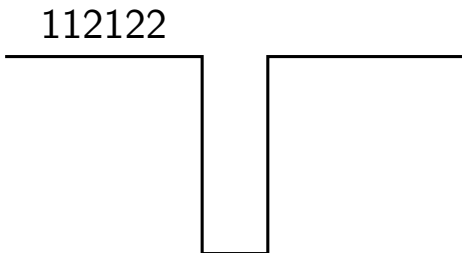
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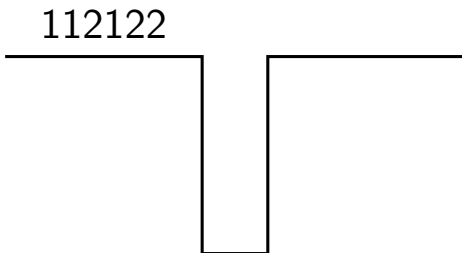
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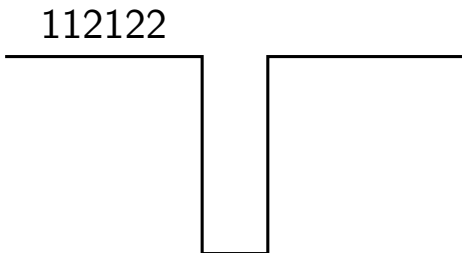


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s_7 “more powerful” than s_6 : every s_6 -sortable word s_7 -sortable.

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s_7 “more powerful” than s_6 : every s_6 -sortable word s_7 -sortable. But:

Theorem (L. 2026)

For all $m \neq m'$, there are words w such that w takes much longer to sort with s_m than $s_{m'}$.